Tanya Goyal

# Research Statement

Large language models (LLMs) are poised to transform how we consume knowledge, work and interact with others. Recent breakthroughs, particularly in their reasoning capabilities, have brought us closer to this future. However, the same models that exhibit Olympiad-level math capabilities also trivially hallucinate facts, struggle with multi-turn conversations, and lack numerous other skills that are typical of humans.

When a lay person asks an LLM a question that would require a human expert to answer (*"Why do transformer architectures use pre-norm?"*) and reads a seemingly detailed and factual response, they naturally assume that the same LLM will excel at queries answerable by non-experts. However, this is not true of today's LLMs that will often hallucinate solutions when asked even simple questions that they cannot answer parametrically. Moreover, the anthropomorphism in the responses (*"I'm sure ..."*, *"I think ..."*, etc.) gives the false impression of human-like abilities of self-reflection and introspection, such as recognizing their own confidence over the generated answer. One direction of my lab's research builds safeguards against such model failures from two angles. First, **we build error detector models** that can post-hoc identify and localize tricky error cases in text generated by modern LLMs [1, 2, 3, 4]. We emphasize localization in our research as it produces fine-grained actionable signals which we successfully use to train inherently more factual LLMs [2, 5]. Second, towards the same goal of error mitigation, we build training strategies and LLMs that, instead of relying on external detectors, learn to self-identify their errors [6, 5, 7]. Specifically, our research produces LLM systems that **can reason over their own limitations and capability boundaries**, and take corrective action during generation. (Part A)

A basic requirement when pursuing modeling improvements is being able to reliably distinguish between meaningful improvements and superficial changes between model versions. Otherwise, we risk spending valuable resources on hill-climbing meaningless or archaic benchmarks. To address this, the second direction of my research focuses on **grounding open-source research on the right metrics**. In addition to building meaningful forward-looking benchmarks [8, 9, 10, 7], we also audit established evaluation frameworks, both human and automatic, to ensure that reported improvements reflect actual progress [11, 12, 13, 14]. (Part B)

## A. Building error detectors and limitation-aware LLMs

As LLMs get integrated into more of our workflows, we will increasingly offload both short- and long-horizon tasks to them. However, today's LLMs lack a crucial ability that even average human personal assistants possess – the ability to calibrate their own capabilities and understand when the assigned task, or a sub-task in a long-horizon task, is outside of their capability boundaries. Today, even the most capable LLMs struggle with abstaining on such queries, and can instead accumulate catastrophic errors in succession. In my lab, we aim to **build not just more capable models, but models that are also aware of the limits of their capabilities**.

**First, why is it challenging to build limitations-aware LLMs?** Most post-training is targeted at improving the accuracy of the final outcome. Therefore, LLMs are incentivized to attempt solving a task even if their probability of success is low. On the other hand, jointly training for the two goals of accuracy and abstention is challenging as there exists a natural trade-off between them.

While similar trade-offs have previously been studied between helpfulness and harmlessness, abstention based on model's own limitations presents unique difficulties. In particular, harmlessness is model-agnostic as, if a query is harmful, all LLMs must abstain. On the other hand, abstention informed by limitation is a model-dependent decision; for instance, stronger and weaker models require abstaining on different sets of questions. Even for the same model, this decision boundary continuously changes during training as capabilities improve.

Our research re-frames limitation-informed abstention as a selective help-seeking problem (**MASH: M**odeling **A**bstention via **S**elective **H**elp-seeking) [6]. Instead of directly training LLMs for abstention, we instead train them to use external search tools to answer complex questions (Figure 1). Firstly, this framing allows us to borrow from recent RL advances that effectively interleave thinking and tool use in reasoning LLMs. Our core idea is to incentivize ***selective help-seeking***, i.e. using external search only when LLMs' own knowledge is lacking, by modifying the standard accuracy-based RL reward to include a search use penalty. As the decision boundary for selective help-seeking, by our definition, is same as that for abstention, our proposed training gives us an abstention model for free! At inference time, we simply remove the trained LLM's access to search and treat any search invocation as an abstention. Our training pipeline also includes a lightweight warm-start step that ensures that the trained LLMs make abstention decisions at the level of atomic facts. MASH outperforms or matches the performance of previous specialized abstention approaches that, unlike MAS, require a-priori information about the LLMs knowledge boundaries. We were particularly encouraged to see improvements over all previous search-augmented baselines when search was enabled during inference, owing to the inductive bias for localization baked into MASH's training.

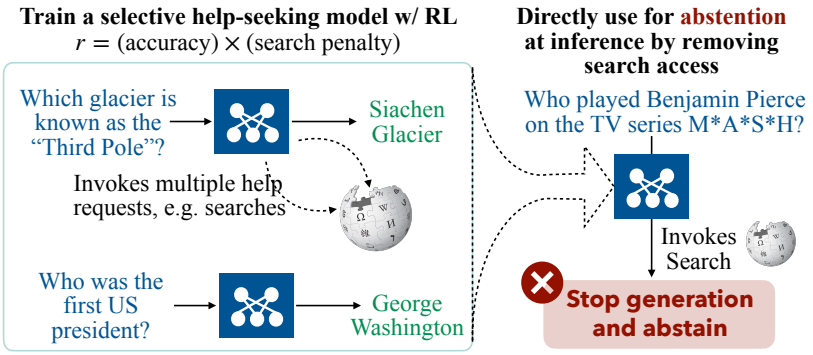


Figure 1: Abstention w/ MASH

**Localizing errors for better modeling and evaluation.** The above research direction contributes to my existing body of work on LLM factuality, which includes building factuality detection models [2, 1, 4], understanding factuality errors of frontier LLMs [2, 15, 11, 9], as well as post-training approaches to improve the factuality of generation models [2, 5, 7]. A common unifying theme in my work is the focus on localization of model errors. I build models and evaluations that localize errors to atomic semantic units of the output. This better informs both downstream analysis work into model limitations as well as subsequent training improvements.

Our factual error detection model for conditional generation, called ***dependency arc entailment*** (DAE), was first in my work to exemplify this principle [1, 2]. DAE decomposes the overall error detection task into smaller entailment tasks that predict whether individual word relationships are entailed by the input (Figure 2). This set of word relationships we evaluated over was informed by the syntactic structure, specifically dependency parse, of the generated text. Our trained DAE model allowed us to predict arc-level factuality and localize errors, e.g. to the *woman* → *Chicago*

arc in Figure 2. It achieved state-of-the-art results on text summarization and paraphrasing when it was first introduced. Furthermore, DAE's error localization was critical in denoising training data [16, 2] to train inherently more factual summarization models. As frontier LLMs became more powerful, we shifted to extracting atomic units via prompting instead of parsing, but combined it with the same localization principle. In all instances, we found that explicitly modeling localized errors results in stronger outcomes compared to response-level error detection.

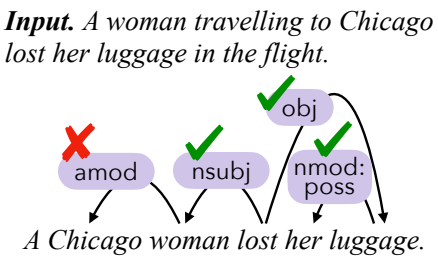

Figure 2: Error localization w/ DAE

**LLM error detectors ⇒ LLMs that self-detect errors.** Despite how strong current LLMs are, guaranteeing factuality remains an open problem. Arguably, even though LLMs now make fewer errors, these errors have an outsized impact because users are more dependant and trusting of LLMs today. Therefore, ensuring factuality remains a key focus of my lab's research. Previously, my work focused on building error detectors for LLM generated text in order to aid evaluation and proper benchmarking. Now, we additionally focus on building LLMs that can model their own epistemic uncertainty and confidence, and self-identify errors. I believe these latter LLMs are critical for real world deployment in long-horizon use cases where we want LLMs to match human experts in both capability and the ability to calibrate their knowledge and capabilities.

### B. Building and auditing evaluation standards

A key aspect of LLM development is reliable evaluation. Due to the rapid pace of development, the LLMs' generation and error spaces change faster than the pace of evaluation metric development. This problem is less pronounced in industry LLM labs that can both hire expensive human annotators during model development and collect feedback from real users after deployment. In contrast, a very common failure mode in academia is the investment of large amounts of resources, both capital and human effort, on hill-climbing archaic metrics that were designed for output distributions that no longer match those of current LLMs. Throughout my research, I have worked to steer evaluation towards real as well as forward-looking system errors.

My research in this direction takes one of two flavors: (1) **ensuring that open-source development is grounded on the right metrics**, where I audit established evaluation frameworks to verify that improvements reported on those actually correspond to meaningful progress [11, 3, 12], and (2) **building evaluations for the next-generation of LLMs**, where I build forward-looking benchmarks that are largely outside of the capability of current frontier LLMs [8, 9, 10] and training techniques [7, 17], and require new paradigms to solve.

**Grounding open-source development on the right metrics.** The goal of automatic metrics, particularly for subjective domains like open-ended generation, is to provide a reasonable proxy for humans. Therefore, metric developers validate their framework against human annotators when proposing a new metric. However, this validation is conducted using state-of-the-art models at the time of proposal even though model behaviors can change dramatically in subsequent development cycles. My work in summarization [11] was the first to convincingly demonstrate this mismatch for models in the post GPT3-era. We showed that established summarization metrics, such as ROUGE

and BERTScore, were no longer valid proxies of humans when comparing models like ChatGPT against much weaker baselines. In fact, we showed that these metrics provide the **opposite** rankings as that of human annotators! Based on our work, similar studies were subsequently conducted for more summarization domains, machine translation, simplification, and others [18, 19, 20, 21, 22].

Since 2022, the lifespan of automatic metrics have become even shorter. With RL-based training, it is increasingly common for trained models to reward-hack evaluation benchmarks instead of making real progress. Our work highlighted another such phenomenon in the field's efforts on reinforcement learning with human feedback (RLHF). RLHF, particularly PPO and its variants, became the favored way of aligning LLMs to human preferences following the success of ChatGPT. However, our work [12] demonstrably proved that most of the reported improvements were superficial – the metric, AlpacaEval, exhibited a bias for longer inputs, RLHF training in open-source domains mainly hill-climbed on this metric by increasing length without improving the quality of the generations itself. This work was extremely impactful, and spurred work in both better length-agnostic metric development work [23, 24] as well as modeling work aimed at improving quality without length-hacking [25, 26]. Throughout my research, I have revisited established evaluation [13, 11, 27], both human and automatic, and even meta-evaluation pipelines [14] to ensure benchmarking standards.

**Benchmarks for the next-generation of LLMs.** To make progress, we need two kinds of benchmarks: immediately useful benchmarks that provide useful signals for training current models, and forward-looking benchmarks that test for capabilities that future models *should* possess. Towards this latter goal, I have collaborated on building challenging benchmarks for long-context modeling [8, 9, 10], retrieval [28], knowledge updating [7], among others. As a representative example, our benchmark, *NoCha* or "***No**vel **Cha**llenge*", evaluates the frontier LLMs ability to verify claims based on novel-length narratives. The best performing LLMs on this benchmark currently lag about $30\%$ points behind humans. Furthermore, these LLMs collectively reported only a $1\%$ improvement in performance in the last 1.5 years since its release. My research contributes such high-quality benchmarks that help us distinguish meaningful progress from incremental ones along target capabilities.

### C. Future Directions

**Building collaborative LLMs** Today, we test LLMs on whether they can perform a task end-to-end, even those like novel-length creative writing which we are unlikely to completely offload to LLMs. This creates a mismatch between test conditions and how we envision using LLMs for long-horizon use cases where users and LLMs will collaboratively create solutions. To serve as effective collaborators, we need LLMs that can proactively probe the user for missing specifications or user preferences, understand and convey gaps in their own capabilities, and learn which sub-steps of a project to offload to stronger collaborators, either humans or other LLMs, amongst numerous other skills. My work on limitation-aware LLMs [6] is a step towards building such reliable LLM collaborators. Going forward, my lab will work on building LLMs that can operate in multi-agent environments by effectively collaborating with other LLMs, or other instances of the same LLM. We will build on our initial work [7] that studied common failure modes of today's solo-trained LLMs when collaborating in the math and coding domains. Additionally, we will build realistic

user models that mimic the evolving knowledge, preferences, and communication styles of users when collaborating with LLMs.

**LLMs that evolve with the evolving world and users** Current LLMs are largely static systems trained on world knowledge up to their training data's cut-off date. However, the world and its facts are constantly evolving. In our prior work [7], we showed that current approaches of continued training on new knowledge corpora allow LLMs to memorize these new facts, but not reason about implicit ways in which they impact the world. How can we train LLMs that not only memorize but reason over new knowledge? Can we build LLMs that actively seek to incorporate new knowledge in their memory, either parametric or non-parametric, when they encounter new concepts in deployment? Our earlier thrust towards building more realistic user models will also help advance research in building LLMs that update behaviors depending on evolving user traits.

## References


[1] **Tanya Goyal** and Greg Durrett. Evaluating factuality in generation with dependency-level entailment. *Findings of EMNLP*, 2020.

[2] **Tanya Goyal** and Greg Durrett. Annotating and modeling fine-grained factuality in summarization. *NAACL*, 2021.

[3] **Tanya Goyal**, Junyi Jessy Li, and Greg Durrett. SNaC: Coherence error detection for narrative summarization. *EMNLP*, 2022.

[4] Ryo Kamoi, **Tanya Goyal**, Juan Diego Rodriguez, and Greg Durrett. Wice: Real-world entailment for claims in wikipedia. *EMNLP*, 2023.

[5] **Tanya Goyal**, Jiacheng Xu, Junyi Jessy Li, and Greg Durrett. Training dynamics for text summarization models. *Findings of ACL 2022*.

[6] Mustafa Omer Gul, Claire Cardie, and **Tanya Goyal**. Pay-per-search models are abstention models. *arXiv*, 2025. In submission to ICLR 2026. Published in NeurIPS 2025 Workshop on Efficient Reasoning.

[7] Aochong Oliver Li and **Tanya Goyal**. Memorization vs. reasoning: Updating LLMs with new knowledge. *Findings of ACL*, 2025.

[8] Yapei Chang, Kyle Lo, **Tanya Goyal**, and Mohit Iyyer. Booookscore: A systematic exploration of book-length summarization in the era of llms. *ICLR*, 2024.

[9] Yekyung Kim, Yapei Chang, Marzena Karpinska, Aparna Garimella, Varun Manjunatha, Kyle Lo, **Tanya Goyal**, and Mohit Iyyer. FABLES: Evaluating faithfulness and content selection in book-length summarization. *COLM*, 2024.

[10] Marzena Karpinska, Katherine Thai, Kyle Lo, **Tanya Goyal**, and Mohit Iyyer. One thousand and one pairs: A "novel" challenge for long-context language models. *EMNLP*, 2024.

[11] **Tanya Goyal**, Junyi Jessy Li, and Greg Durrett. News summarization and evaluation in the era of GPT-3. *arXiv*, 2022.

[12] Prasann Singhal, **Tanya Goyal**, Jiacheng Xu, and Greg Durrett. A long way to go: Investigating length correlations in rlhf. *COLM*, 2024.

[13] Wenting Zhao, Alexander M Rush, and **Tanya Goyal**. Challenges in trustworthy human evaluation of chatbots. *Findings of NAACL*, 2025.

[14] Tianruo Rose Xu, Vedant Gaur, Liu Leqi, and **Tanya Goyal**. The progress illusion: Revisiting meta-evaluation standards of llm evaluators. *Findings of EMNLP*, 2025.

[15] Liyan Tang, **Tanya Goyal**, Alexander R Fabbri, Philippe Laban, Jiacheng Xu, Semih Yahvuz, Wojciech Kryściński, Justin F Rousseau, and Greg Durrett. Understanding factual errors in summarization: Errors, summarizers, datasets, error detectors. *ACL*, 2023.

[16] Yanzhu Guo, Chloé Clavel, Moussa Kamal Eddine, and Michalis Vazirgiannis. Questioning the validity of summarization datasets and improving their factual consistency. *EMNLP*, 2022.

[17] Aochong Oliver Li and **Tanya Goyal**. Off-Trajectory Reasoning: Can LRMs Collaborate on Reasoning Trajectory? *arxiv*, 2025. In submission to ICLR. Published in NeurIPS 2025 Workshop on Efficient Reasoning.

[18] Sebastian Joseph, Lily Chen, Jan Trienes, Hannah Göke, Monika Coers, Wei Xu, Byron Wallace, and Junyi Jessy Li. FactPICO: Factuality evaluation for plain language summarization of medical evidence. *ACL*, 2024.

[19] Lucy Lu Wang, Julia Otmakhova, Jay DeYoung, Thinh Hung Truong, Bailey Kuehl, Erin Bransom, and Byron C Wallace. Automated metrics for medical multi-document summarization disagree with human evaluations. 2023.

[20] Tianyi Zhang, Faisal Ladhak, Esin Durmus, Percy Liang, Kathleen McKeown, and Tatsunori B Hashimoto. Benchmarking large language models for news summarization. *TACL*, 2024.

[21] Xiao Pu, Mingqi Gao, and Xiaojun Wan. Summarization is (almost) dead. *arXiv preprint arXiv:2309.09558*, 2023.

[22] Chantal Shaib, Millicent Li, Sebastian Joseph, Iain Marshall, Junyi Jessy Li, and Byron Wallace. Summarizing, simplifying, and synthesizing medical evidence using GPT-3 (with varying success). 2023.

[23] Yann Dubois, Percy Liang, and Tatsunori Hashimoto. Length-controlled alpacaeval: A simple debiasing of automatic evaluators. *COLM*, 2024.

[24] Nathan Lambert, Valentina Pyatkin, Jacob Morrison, Lester James Validad Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, et al. Rewardbench: Evaluating reward models for language modeling. *Findings of NAACL*, 2025.

[25] Ryan Park, Rafael Rafailov, Stefano Ermon, and Chelsea Finn. Disentangling length from quality in direct preference optimization. *Findings of ACL*, 2024.

[26] Junru Lu, Jiazheng Li, Siyu An, Meng Zhao, Yulan He, Di Yin, and Xing Sun. Eliminating biased length reliance of direct preference optimization via down-sampled kl divergence. *EMNLP*, 2024.

[27] Zhiyuan Zeng, Jiatong Yu, Tianyu Gao, Yu Meng, **Tanya Goyal**, and Danqi Chen. Evaluating large language models at evaluating instruction following. *ICLR*, 2024.

[28] Anirudh Ajith, Mengzhou Xia, Alexis Chevalier, **Tanya Goyal**, Danqi Chen, and Tianyu Gao. Litsearch: A retrieval benchmark for scientific literature search. *EMNLP*, 2024.